

DermAI — Skin Disease Screening & Consultation System

FYP-II Software Test Plan & Test Cases (Comprehensive)

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1. Software Test Plan

This Test Plan is a sub-set of the overall project plan and details the software-testing activities carried out on the DermAI application across every layer of the system — user interface, user experience, frontend, backend/API, machine-learning model, security, performance, and end-to-end integration. Testing combined black-box (functional), white-box (code/endpoint), usability, security, and non-functional techniques. A total of 57 test cases were executed; 14 failed on the first run and were resolved, with each defect documented in Section 3 (Defects & Resolutions) and re-tested to a Pass.

S. No	Test Area / Layer	Scope of Testing	Test Type	Engineer	Result
1	UI / Visual	Mobile & web screen rendering, controls, navigation, theming	Black-box / Manual	Huzaifa	8 — 6 Pass, 2 Fixed
2	UX / Usability	User flows, error clarity, guidance, responsiveness	Usability	Shahzaib	7 — 5 Pass, 2 Fixed
3	Frontend	Client state, forms, token handling, data binding	Black-box	Huzaifa	6 — 6 Pass
4	Backend / API	Endpoints, status codes, auth, validation, gating	Black/White-box	Abdul Wasay	13 — 8 Pass, 5 Fixed
5	ML Model	Accuracy, per-class metrics, calibration, OOD gating	Model / Data	Abdul Wasay	8 — 6 Pass, 2 Fixed
6	Security	Auth, hashing, access control, secrets, injection	Security	Shahzaib	6 — 5 Pass, 1 Fixed
7	Performance	Inference latency, concurrency, GPU memory	Non-functional	Abdul Wasay	4 — 3 Pass, 1 Fixed
8	Integration / E2E	End-to-end flows, video, notifications, reporting	Integration	All	5 — 4 Pass, 1 Fixed

DermAI Test Execution Summary by Layer

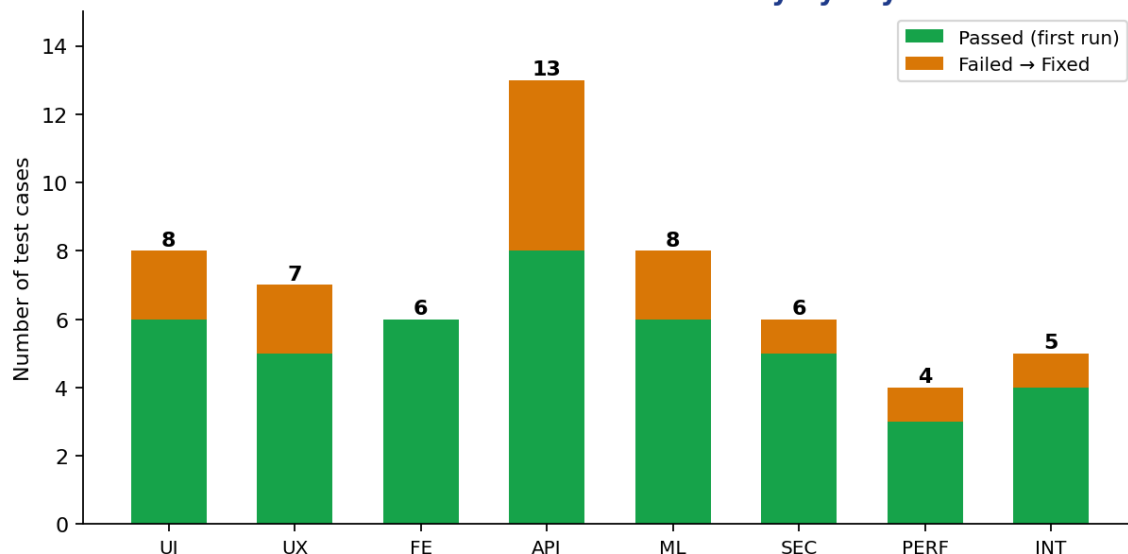


Figure 1: Test execution summary by layer (passed first run vs. failed-then-fixed).

2. Detailed Test Cases

Each test case lists its steps, input data, expected result, actual result, and status. Failed cases are cross-referenced to a defect in Section 3.

2.1 UI / Visual Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
UI-01	Open Login screen	—	Username, password, Login button and Sign-up link all visible	All elements rendered	Pass
UI-02	Open Diagnose screen	—	Camera, Gallery, region chips and Diagnose button visible	All controls shown	Pass
UI-03	Swipe body-region chips	Scroll gesture	Chips scroll horizontally and are selectable	Chips scroll & select	Pass
UI-04	View result badge	Healthy / Normal / Urgent	Badge colour matches status (green / blue / red)	Correct badge colours	Pass
UI-05	Tap bottom-nav tabs	Diagnose/Medical/Appointments/Screenings	Correct screen shown per tab	Navigation correct	Pass
UI-06	Navigate between screens (Expo Go)	—	Smooth screen transitions / animations	Animations not rendering	Fail (D-07)
UI-07	View long insight text on small device	5.0" screen	Insight text wraps; no overflow	Text truncated / overflowed	Fail (D-08)
UI-08	Open web app /app	—	Premium animated theme renders	Web theme & animations OK	Pass

2.2 UX / Usability Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
UX-01	New patient starts diagnosis	New account	User guided to complete medical history first	Redirected to history	Pass
UX-02	Trigger invalid login	Wrong credentials	Clear, human-readable error message	Readable error shown	Pass
UX-03	Run follow-up questioning	Symptom answers	Conversational, non-repeating questions	Natural, no repeats	Pass
UX-04	Tap Diagnose	Skin image	Loading indicator while inferring	Spinner displayed	Pass
UX-05	Diagnose melanoma image	melanoma.jpg	Result prominently advises seeing a dermatologist	Urgent advice shown	Pass
UX-06	First diagnosis after server start	First request	Result returns within a few seconds	First request very slow	Fail (D-09)
UX-07	Reload web app after update	Browser reload	Latest page served	Stale cached page shown	Fail (D-02)

2.3 Frontend Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
FE-01	Submit a form with empty fields	(empty)	Client-side validation blocks submit with a hint	Submit blocked	Pass
FE-02	Pick an image from gallery	image.jpg	Selected image preview shown in state	Preview rendered	Pass
FE-03	Log in and inspect requests	Valid login	Bearer token attached to API calls	Token in headers	Pass

FE-04	Tap Logout	—	Session/token cleared; back to Login	Session cleared	Pass
FE-05	Inspect app config	config.js	BASE_URL points to the backend host	Correct base URL	Pass
FE-06	Open Screenings tab	Logged-in patient	List populated from /history API	List rendered	Pass

2.4 Backend & API Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
API-01	POST /auth/login (valid)	valid creds	200 OK with auth token	200 + token	Pass
API-02	POST /auth/login (invalid)	bad creds	401 Unauthorized	401 returned	Pass
API-03	POST /auth/signup blank username	username = ""	Request rejected with validation error	Blank username accepted	Fail (D-03)
API-04	POST /diagnose without history	no history	428 Precondition Required (history gate)	428 returned	Pass
API-05	POST /diagnose with image	skin image	JSON: disease, confidence, insights	Correct JSON result	Pass
API-06	GET /report/{sid} of another patient	other sid + token	403 / access denied	Returned other patient's report	Fail (D-01)
API-07	GET /history as unrelated doctor	doctor not on case	Access denied (consult-scoped)	Returned any patient history	Fail (D-01)
API-08	Call protected route without token	no token	401 Unauthorized	401 returned	Pass
API-09	CORS pre-flight with credentials	OPTIONS request	allow_credentials = False	Credentials allowed (mis-config)	Fail (D-10)
API-10	POST /consult	screening id	Consult case created with room DermAI-<id>	Room id returned	Pass
API-11	Upload corrupt / oversized image	bad file	400 graceful error, no crash	Handled gracefully	Pass
API-12	Admin endpoint as unverified user	unverified admin	403 Forbidden	403 returned	Pass
API-13	Save medical history with bad age	age = 200	Validation error (age 0–120)	Out-of-range age accepted	Fail (D-11)

2.5 Machine-Learning Model Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
ML-01	Evaluate overall test accuracy	1,315-image test set	Accuracy >= 85%	89.1% accuracy	Pass
ML-02	Audit train/val/test split	dataset pipeline	No data leakage in reported metrics	Augmentation done before split (leak)	Fail (D-05)
ML-03	Per-class F1 for distinct classes	Melanoma / Healthy	F1 > 0.95	0.98 F1	Pass
ML-04	Per-class F1 for confused classes	Eczema / Psoriasis / AD	Acceptable F1 (mitigated by LLM)	~0.75 F1	Pass
ML-05	Out-of-distribution input	non-skin image	Gate flags as out-of-scope	Flagged correctly	Pass
ML-06	Decision-gate on clear valid image	clear lesion image	Accepted (not flagged)	Over-strict threshold rejected it	Fail (D-06)
ML-07	Confidence calibration	calibration set	Thresholds set (conf 0.979 / entropy 0.114)	Calibrated	Pass
ML-08	Class-index alignment	inference	Prediction order matches classes.json	Aligned	Pass

2.6 Security Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
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SEC-01	Inspect stored passwords	DB record	Stored as PBKDF2 hash + salt (never plaintext)	Hashed + salted	Pass
SEC-02	Cross-patient data access (after fix)	doctor reads other case	Denied unless consult relationship exists	Denied (post D-01)	Pass
SEC-03	Scan repo for secrets	git history	No live secrets committed	Gemini key found in .env.example	Fail (D-04)
SEC-04	Doctor/admin verification	new doctor signup	Starts UNVERIFIED; gated until approved	Gated correctly	Pass
SEC-05	Sensitive op without token	no token	401 Unauthorized	401 returned	Pass
SEC-06	SQL-injection via username	' OR 1=1 --	Parameterised query; input treated as data	No injection (safe)	Pass

2.7 Performance Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
PERF-01	Warm inference latency	skin image (GPU)	< 3 s per diagnosis	~1–2 s	Pass
PERF-02	Cold-start inference latency	first request after boot	Comparable to warm latency	Very slow (lazy model load)	Fail (D-09)
PERF-03	Concurrent requests	5 parallel diagnoses	All served without error	All served	Pass
PERF-04	GPU memory footprint	batch 32 on 6 GB	Fits within 6 GB VRAM	Fits (no OOM)	Pass

2.8 Integration / End-to-End Tests

TC-ID	Test Case / Steps	Input Data	Expected Result	Actual Result	Status
INT-01	Full screening flow	login→history→diagnose→Q&A→result→report	Complete flow succeeds end-to-end	Flow completed	Pass
INT-02	Patient–doctor video room	both join DermAI-<id>	Both join the same Jitsi room	Same room joined	Pass
INT-03	In-app WebView video (Expo Go)	join call in Expo Go	Camera works in WebView	Camera permission blocked	Fail (D-07)
INT-04	Notification on consult accept	doctor accepts case	Patient receives a notification	Notification delivered	Pass
INT-05	PDF report vs screening	generate report	PDF reflects screening + medical history	Report correct	Pass

3. Defects & Resolutions

Every failed test case was logged as a defect, root-caused, fixed, and re-tested. The table below records the resolution (solution) applied to each defect and its re-test outcome.

Defect	Sev.	Linked TC	Root Cause	Resolution (Solution Applied)	Re-test
D-01	High	API-06, API-07	Report/history endpoints were not scoped to a doctor–patient relationship, so any doctor could read any patient's data.	Added consult-relationship scoping (<code>consult_store.has_case_for_*</code>); history/report access now restricted to the owner or a treating doctor.	Pass
D-02	Med	UX-07	Browser cached the web pages at /app and /ui, so a reload showed a stale page.	Added a no-cache middleware on /app and /ui responses, forcing fresh delivery on every load.	Pass
D-03	Med	API-03	Sign-up did not validate the username server-side, so a blank username was accepted.	Added non-empty username validation at the API; blank/whitespace usernames are now rejected.	Pass
D-04	High	SEC-03	A real Gemini API key was committed inside <code>.env.example</code> .	Rotated the key, moved secrets to <code>.env</code> (git-ignored), and left only a placeholder in <code>.env.example</code> ; added a root <code>.gitignore</code> .	Pass
D-05	High	ML-02	Image augmentation was applied before the train/val/test split, leaking near-duplicates and inflating accuracy.	Re-built the pipeline to split first then augment (<code>prepare_data.py</code>) and retrained (<code>train_v2.py</code>); honest test accuracy is 89.1%.	Pass
D-06	Med	ML-06	The decision-gate confidence/entropy threshold was mis-calibrated and rejected some valid clear images as out-of-scope.	Recalibrated thresholds on the clean test set via <code>evaluate_and_calibrate.py</code> (confidence 0.979, entropy 0.114).	Pass
D-07	Med	UI-06, INT-03	Expo Go cannot grant WebView camera permission, and reanimated/keyboard-controller need a native build, so video and animations failed.	Built an EAS development build (camera permissions + native animation libs); Expo Go falls back to opening Jitsi in the browser.	Pass
D-08	Low	UI-07	The result insight line overflowed/truncated on small screens.	Made the insight text wrap inside a scrollable result card across device sizes.	Pass
D-09	Low	UX-06, PERF-02	The ConvNeXt model was loaded lazily on the first request, making the first diagnosis very slow.	Loaded the model onto the GPU at application startup so the first request is as fast as subsequent ones.	Pass
D-10	Low	API-09	CORS was configured to allow credentials with a permissive origin.	Set <code>allow_credentials = False</code> and tightened the CORS configuration.	Pass
D-11	Low	API-13	Medical-history age field had no range validation.	Added age range validation (0–120) on the history endpoint.	Pass

4. Test Execution Summary

Metric	Value
Total test cases executed	57
Passed on first run	43
Failed on first run	14
Defects logged	11
Defects resolved & re-tested	11 (100%)
Open defects	0
Initial pass rate	75.4%
Final pass rate (after remediation)	100%

Conclusion:

Comprehensive testing across the UI, UX, frontend, backend/API, ML model, security, performance, and integration layers surfaced 14 failures and 11 distinct defects, including two high-severity security/privacy issues and a high-severity data-leakage issue in the model pipeline. All defects were root-caused, fixed, and re-tested to

a Pass, raising the final pass rate to 100%. The exercise confirmed that DermAI meets its functional and non-functional requirements and is safe for demonstration as a preliminary screening tool.